

D782

Network Architecture and Cloud Computing

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A 40-person accounting firm wants email and file sharing without running servers or patching software. The firm's partners prefer familiar tools. Which cloud service model fits?

- A. Private cloud.
- B. IaaS.
- C. SaaS.
- D. PaaS.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A development team wants to deploy its own application code without managing the operating system, patches, or runtime. Which service model fits?

- A. PaaS.
- B. On-premises servers.
- C. SaaS.
- D. IaaS.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A 350-bed hospital keeps patient records in its own data center for regulatory reasons but moves its public website's extra traffic to a public cloud during flu season. Which deployment model is this?

- A. Private cloud.
- B. Hybrid cloud.
- C. Public cloud.
- D. Community cloud.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An online store needs 10 servers for 25 days a month and 40 servers for 5 sale days. Each server costs \$5 per day. The store's logo was redesigned last spring. How much does it save each month by scaling elastically instead of running 40 servers all month?

- A. \$4,500
- B. \$2,250
- C. \$6,000
- D. \$3,750

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A workload costs \$0.40 per hour on demand. A one-year reserved commitment costs \$2,190. Below how many hours of use per year is on demand cheaper?

- A. 5,475 hours
- B. 8,760 hours
- C. 547.5 hours
- D. 876 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A company runs virtual machines on an IaaS provider. A storage bucket holding customer files was set to public read, and the files were exposed. The provider holds major security certifications. Who was responsible for the setting?

- A. The customer.
- B. The provider, because it is certified.
- C. No one; cloud storage is always public by default.
- D. Both equally.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

In a SaaS arrangement, which task remains the customer's responsibility?

- A. Nothing at all.
- B. Patching the application.
- C. Physical security of the data centers.
- D. Managing user accounts, access permissions, and the data it stores.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A company with three offices hosts a public web server and an internal HR database. Where should each be placed?

- A. Both on the internal network.
- B. No firewall needed if the site uses HTTPS.
- C. Web server in a DMZ between firewalls; HR database on the protected internal network.
- D. Both in the DMZ.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A remote-access policy at a 600-employee company requires a password and an answer to a security question. The IT team calls this multifactor authentication. Is it?

- A. No; MFA always requires biometrics.
- B. No; both are the same factor type.
- C. Yes; it uses two items.
- D. Yes, for companies with more than 500 employees.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which pairing is true multifactor authentication?

- A. Two different passwords.
- B. A PIN and a password.
- C. A password and a one-time code from an authenticator app on the user's phone.
- D. A fingerprint and a face scan.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A factory's camera system must reject defective parts within 20 ms, but the nearest cloud region is 60 ms away round trip. The factory makes 5,000 parts an hour. Which architecture fits?

- A. Slow production below 5,000 parts an hour.
- B. Run the inspection model on edge devices at the factory and send summaries to the cloud.
- C. Keep everything in the cloud and buy more bandwidth.
- D. Use larger cloud instances.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A shopping-cart service runs in 12 regions. It must keep accepting items even when the network link between regions fails, and brief inconsistencies are acceptable. Under the CAP theorem, which trade-off is being made?

- A.** Avoid partitions entirely to get both.
- B.** Favor consistency over availability.
- C.** Reduce to fewer than 12 regions.
- D.** Favor availability over strong consistency during partitions.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A 40-person accounting firm wants email and file sharing without running servers or patching software. ~~The firm's partners prefer familiar tools.~~ Which cloud service model fits?

Target word: service model **Decides:** They want finished software with nothing to manage.

A	Confusion set	A deployment model, not a service model.
B	Inverted operation	Leaves the most management to the customer.
C	Correct	The provider runs and maintains the application.
D	Adjacent category	For running your own code, not using finished apps.

2. Answer A Concept 1 • transfer

A development team wants to deploy its own application code without managing the operating system, patches, or runtime. Which service model fits?

Target word: service model — reversed need **Decides:** Own code, managed platform.

A	Correct	The platform is managed; the team supplies code.
B	Confusion set	Maximum self-management.
C	Inverted operation	Finished software; no place for the team's own code.
D	Adjacent category	The team would manage the OS and runtime.

3. Answer B Concept 2

~~A 350-bed~~ hospital keeps patient records in its own data center for regulatory reasons but moves its public website's extra traffic to a public cloud during flu season. Which deployment model is this?

Target word: deployment model **Decides:** Private plus public, working together.

A	Adjacent category	Drops the public bursting.
B	Correct	Combines private and public with workloads moving between them.
C	Dropped element	Drops the private data center.
D	Confusion set	Shared by organizations with common needs; not described.

4. Answer D Concept 3

An online store needs 10 servers for 25 days a month and 40 servers for 5 sale days. Each server costs \$5 per day. ~~The store's logo was redesigned last spring.~~ How much does it save each month by scaling elastically instead of running 40 servers all month?

Target word: monthly savings **Decides:** Elastic = $(250 + 200) \times \$5 = \$2,250$; fixed = $40 \times 30 \times \$5 = \$6,000$.

A	Confusion set	Compares against a 10-server fleet ($\$6,000 - \$1,500$), which cannot cover the sale.
B	Adjacent category	The elastic cost, not the savings.
C	Dropped element	The fixed-fleet cost, with elastic cost left out.
D	Correct	$\$6,000 - \$2,250 = \$3,750$.

5. Answer A Concept 3 · transfer

A workload costs \$0.40 per hour on demand. A one-year reserved commitment costs \$2,190. Below how many hours of use per year is on demand cheaper?

Target word: solve for break-even hours **Decides:** Hours = \$2,190 / \$0.40.

A	Correct	$2,190 / 0.40 = 5,475$.
B	Adjacent category	The hours in a year, not the break-even point.
C	Wrong stem number	Misreads the rate as \$4.00 per hour.
D	Inverted operation	Multiplies $2,190 \times 0.40$.

6. Answer A Concept 4

A company runs virtual machines on an IaaS provider. A storage bucket holding customer files was set to public read, and the files were exposed. ~~The provider holds major security certifications.~~ Who was responsible for the setting?

Target word: responsibility **Decides:** Access configuration is the customer's job.

A	Correct	Customers own data and access configuration under shared responsibility.
B	Camouflage	Certifications cover the provider's layer, not customer settings.
C	Absolute language	Access is set by the customer.
D	Adjacent category	Responsibility is split by layer, not shared evenly for every setting.

7. Answer D Concept 4 · transfer

In a SaaS arrangement, which task remains the customer's responsibility?

Target word: customer duty in SaaS — reversed **Decides:** Customers always own identities and data.

A	Absolute language	Customers still own access and data.
B	Inverted operation	The SaaS provider patches.
C	Confusion set	Always the provider's layer.
D	Correct	Identity and data stay with the customer in every model.

8. Answer C Concept 5

A company ~~with three offices~~ hosts a public web server and an internal HR database. Where should each be placed?

Target word: network placement **Decides:** Public-facing vs. sensitive internal.

A	Dropped element	Drops the buffer zone for the public server.
B	Confusion set	Encryption in transit is not access control.
C	Correct	Limits exposure if the public server is compromised.
D	Adjacent category	Exposes the sensitive database.

9. Answer B Concept 6

A remote-access policy ~~at a 600-employee company~~ requires a password and an answer to a security question. The IT team calls this multifactor authentication. Is it?

Target word: MFA check **Decides:** Both are something you know.

A	Absolute language	A possession factor also works.
B	Correct	MFA needs different factor types.
C	Dropped element	Drops the factor-type rule.
D	Camouflage	Headcount does not define MFA.

10. Answer C Concept 6 · transfer

Which pairing is true multifactor authentication?

Target word: valid MFA — reversed **Decides:** Two different factor types.

A	Inverted operation	Adds more of the same factor, not a new type.
B	Confusion set	Both are knowledge.
C	Correct	Something you know plus something you have.
D	Adjacent category	Both are inherence.

11. Answer B Concept 7

A factory's camera system must reject defective parts within 20 ms, but the nearest cloud region is 60 ms away round trip. ~~The factory makes 5,000 parts an hour.~~ Which architecture fits?

Target word: architecture choice **Decides:** The round trip alone exceeds the deadline.

A	Camouflage	Volume does not decide per-part latency.
B	Correct	Local processing meets the latency limit.
C	Dropped element	Bandwidth does not cut latency below 60 ms.
D	Adjacent category	Compute speed does not shorten network round trips.

12. Answer D Concept 8

A shopping-cart service ~~runs in 12 regions~~. It must keep accepting items even when the network link between regions fails, and brief inconsistencies are acceptable. Under the CAP theorem, which trade-off is being made?

Target word: CAP choice **Decides:** Keep serving during partitions; tolerate inconsistency.

A	Absolute language	Partitions cannot be ruled out across regions.
B	Inverted operation	That would reject writes during a partition.
C	Camouflage	Region count does not remove the trade-off.
D	Correct	The cart stays writable and reconciles later.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Cloud service models	IaaS gives virtual infrastructure you manage from the OS up; PaaS runs your code on a managed platform; SaaS delivers finished software.	Confusing service models with deployment models, or IaaS with SaaS.	How much the customer wants to manage.
2	Cloud deployment models	Public = shared provider cloud; private = one organization; community = shared by organizations with common needs; hybrid = combination with workloads moving between.	Calling a mix of on-premises and public cloud either one alone.	Sensitive data kept in-house with bursts to a public provider.
3	Elasticity and cloud cost	Pay for capacity as used; compare total elastic cost with fixed peak capacity or reserved pricing.	Comparing against the wrong baseline or reporting cost as savings.	Server counts by days, or on-demand vs. reserved rates.
4	Shared responsibility	The provider secures the cloud's underlying infrastructure; the customer always secures its data, identities, and configurations, and more as it moves toward IaaS.	Assuming the provider handles customer misconfigurations.	A misconfigured storage bucket or access setting.
5	Perimeter security and the DMZ	Public-facing servers go in a DMZ between firewalls; internal systems stay on the protected internal network.	Putting public servers inside with sensitive data, or relying on encryption alone.	A public web server and an internal database.
6	Multifactor authentication	MFA combines different factor types: something you know, have, or are.	Counting two items of the same type as MFA.	Password plus security question.
7	Edge computing	Process data near its source when latency, bandwidth, or connectivity make round trips to the cloud impractical.	Trying to fix latency with bigger cloud servers.	A millisecond deadline shorter than the cloud round trip.
8	CAP trade-off in distributed systems	During a network partition a distributed system must favor consistency or availability; partitions cannot be ruled out.	Claiming to avoid partitions (CA) across regions.	"Keep accepting writes when regions lose contact."

Course scope (public description): In this course, students are equipped to understand, analyze, and implement cloud computing solutions tailored to diverse business needs. The curriculum offers an in-depth look at state-of-the-art cloud technologies, the underlying business trends fueling cloud adoption, and the foundational systems of modern data center computing. Participants will learn to identify and apply various cloud systems to specific business cases, understand the interplay between network systems to form cloud computing environments, and utilize cloud computing models to address infrastructure challenges. A focus on network security enables learners to assess and enhance network security systems, covering cybersecurity principles, perimeter security, authentication, and the creation of secure networks. The course also delves into scalable distributed systems and programming frameworks that facilitate edge computing, illustrating how cloud-based solutions can resolve real-world business problems.